# Impact energy and TNT equivalents

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Convert an energy estimate without claiming a complete damage prediction.

**SUCCESS CRITERIA - I can...**

1. Convert speed units before squaring.
2. Calculate kinetic energy from the supplied mass and velocity.
3. Distinguish an energy equivalent from an explosion or damage forecast.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “equivalent” and “kinetic energy” to explain your answer. Sentence starters are optional.

**equivalent**: equal in the specified quantity, not necessarily in other effects

**kinetic energy**: energy associated with motion

**impact**: a collision or strike between objects

## 01 Warm-up: bring back an earlier idea

A constant **30 N** forward force acts at steady **5 m/s** for **7 s**. **Find mechanical power** and work.

**Word help:** power = rate of energy transfer

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $K = \frac{1}{2}mv^2$
TNT equivalent (tons) = K/(**$4.184\times10^9$ J**).
**1 ton** TNT equivalent = **$4.184\times10^9$ J**. Equal energy does not imply identical effects.

**Worked reasoning:** A fictional event transfers **$8.4\times10^6$ J**. Using **$4.2\times10^6$ J** per kg TNT as an energy comparison gives **2 kg** TNT equivalent. This compares energy, not identical effects.

**Your turn:** Draw a labeled energy comparison and **name one effect this number** alone cannot predict.

**Sentence starter:** I predict ... . I used ... to make this prediction because ... .

**Initial** **Transfers** **Final**

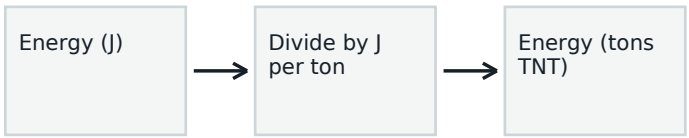
Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

Two reports describe the same fictional impact: one uses joules and the other uses TNT-equivalent tons.
1. Use the conversion diagram to **explain why the units can** express the same quantity.
2. What does this comparison leave out?

**Sentence starter:** For this answer, I treated ... as true. I found ... . This supports ... because ... .




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

A hypothetical **$10^6$ kg** asteroid moves at **10 km/s**. Convert speed to m/s, calculate kinetic energy and express that energy in TNT-equivalent tons using **$4.184 \times 10^9$ J** per ton. **Complete the conversion table**.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

| Energy (J) | J per ton TNT | Equivalent tons |
|---|---|---|
| ____ | $4.184 \times 10^9$ | ____ |

**Marzano check - circle:** ① ② ③

# Impact energy and TNT equivalents

**antocciphysics**



Continue down each column.

Sentence starters are optional. You may plan in any language.

## 05 Explain your reasoning

Why should an energy-equivalent number not be reported as a precise area of destruction?

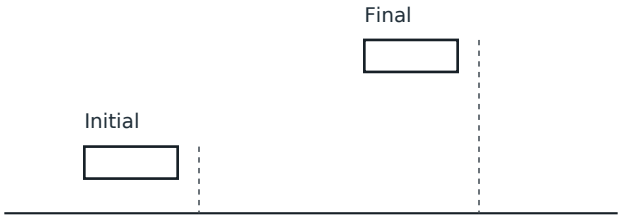
**Sentence starter:** For this answer, I treated … as true. I found … . This supports … because … .

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

A **5 kg** object descends **7 m** near Earth. Use g=**10**. Find ΔU and the work done by gravity.
**Label the distances or energy** levels used in your reasoning.

**Sentence starter:** I found … [unit]. This value means … in this situation.




Choose and label a zero-height level.

**Marzano check - circle:** ① ② ③

## 07 Exit check

At fixed speed, mass doubles while assumed energy coupling fraction halves.
1. **Compare coupled energy**.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

**Sentence starter:** Both … . They differ in … because … .

**Initial** **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the prediction

A **2000 kg** hypothetical object moves at **1000 m/s**.
1. **Find kinetic energy** and kg TNT equivalent using **4.2×10⁶ J/kg**.
2. **Explain why neither alone predicts** a crater size.
3. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

**Sentence starter:** Both … . They differ in … because … .

Total energy → Area / coupling → Local effects

**Marzano check - circle:** ① ② ③